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Technical Specification

O-RAN White Box Hardware Working Group
Deployment Scenarios and Base Station Classes

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Chapter 1 Introductory Material

1.1 Scope

This Technical Specification has been produced by the O-RAN.org.

The contents of the present document are subject to continuing work within O-RAN WG7 and may change following formal O-RAN approval. Should the O-RAN.org modify the contents of the present document, it will be re-released by O-RAN Alliance with an identifying change of release date and an increase in version number as follows:

Release x.y.z

where:

x the first digit is incremented for all changes of substance, i.e., technical enhancements, corrections, updates, etc. (the initial approved document will have x=01).

y the second digit is incremented when editorial only changes have been incorporated in the document.

z the third digit included only in working versions of the document indicating incremental changes during the editing process. This variable is for internal WG7 use only.

The present document specifies operator requirements for scenarios, use cases and base station classes for O-RAN white box hardware, and identifies typical deployment scenarios associated with carrier frequency, inter-site distance, and other key parameters and attributes.

In the main body of this specification (in any "chapter") the information contained therein is informative, unless explicitly described as normative. Information contained in an "Annex" to this specification is always informative unless otherwise marked as normative.

1.2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document.

[1] 3GPP TR 21.905, "Vocabulary for 3GPP Specifications"
[2] 3GPP TR 38.104, "NR; Base Station (BS) radio transmission and reception"
[3] 3GPP TR 38.473, "NG-RAN F1 Application Protocol (F1AP)"
[4] 3GPP TR 38.340, "NR; Backhaul Adaptation Protocol"
[5] 3GPP TR 38.874, "Study on Integrated Access and Backhaul"
[6] O-RAN.WG7.CUS.0-v02.00, O-RAN WG4 Control, User and Synchronization Specification

1.3 Definitions and Abbreviations

1.3.1 Definitions

For the purposes of the present document, the terms and definitions given in [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in [1]. For the base station classes of Pico, Micro and Macro, the definitions are given in [2].

All-In-One architecture: In the all-in-one architecture, the O-RU, O-DU and O-CU are implemented on one platform. There is no need for neither fronthaul interface between O-RU and O-DU nor F1 interface between O-DU and O-CU.

Carrier Frequency: Central frequency of the cell.

F1 interface: The open interface between O-CU and O-DU, the definition is given in O-RAN WG5. Refer to [3].

IAB-donor: RAN node which provides UE's interface to core network and wireless backhauling functionality to IAB nodes. Refer to [4] and [5].

IAB-node: AN node that supports wireless access to UEs and wirelessly backhauls the access traffic. Refer to [4] and [5].

Integrated architecture: In the integrated architecture, the O-RU and O-DU are implemented on one platform. Each O-RU and RF frontend is associated with one LLS-DU. They are then aggregated to O-CU, connected by F1 interface.

Layout: Network framework.

Mobile-Termination (MT): MT is the UE function in each IAB node that allows it to communicate with parent node. Refer to [4] and [5].

Split architecture: The O-RU and O-DU are physically separated from one another in this architecture. A fronthaul gateway may aggregate multiple O-RUs to one O-DU. O-DU, fronthaul gateway and O-RUs are connected by fronthaul interfaces.

System bandwidth: The bandwidth in which a Base Station transmits and receives multiple carriers and/or RATs simultaneously.

Transmission Reception Point (TRxP): Antenna array with one or more antenna elements available to the network located at a specific geographical location for a specific area.

Layer: This term refers to deployment layers. A single layer has no overlay cell. Two layers refers to a layout where the second layer maybe deployed as underlaying cell w.r.t the primary cell.

1.3.2 Abbreviations

For the purposes of the present document, the abbreviations given in [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in [1].

7-2x	Fronthaul interface split option as defined by O-RAN WG4
3GPP	Third Generation Partnership Project
5G	Fifth-Generation Mobile Communications
AC	Alternating Current
BB	Baseband
BDS	BeiDou Navigation Satellite System
bps	Bits Per Second
BS	Base Station
CU	Centralized Unit as defined by 3GPP
DC	Direct Current
DL	Downlink
DRX	Discontinuous Reception
DU	Distributed Unit as defined by 3GPP
EDGE	Enhanced Data rates for GSM Evolution
EIRP	Equivalent Isotropic Radiated Power
eMBB	Enhanced Mobile Broadband
FDD	Frequency Division Duplex
FH	Fronthaul
$FHGW_{x\rightarrow y}$	Fronthaul gateway that can translate FH protocol from an O-DU with split option x to an O-RU with split option y, with currently available option 7-2x→8.
FHM	Fronthaul gateway with no FH protocol translation, supporting an O-DU with split option x and an O-RU with split option x, with currently available options 6→6, 7-2x→7-2x and 8→8.
Gbps	Gigabits Per Second
GNSS	Global Navigation Satellite System
GSM	Global System for Mobile communications
HAR	Hardware Architecture and Requirement
HRD	Hardware Reference Design
IAB	Integrated Access and Backhaul
IEEE	Institute of Electrical and Electronics Engineers
ISD	Inter-Site Distance
KPI	Key Performance Indicator
MAC	Medium Access Control
MEC	Mobile/Multi-access Edge Computing
MIMO	Multiple Input Multiple Output

Ms	Milliseconds
MT	Mobile-Termination
MU-MIMO	Multiple User MIMO
NB-IoT	Narrow Band Internet of Things
NR	New Radio
O-CU	O-RAN Centralized Unit as defined by O-RAN
O-DU	O-RAN Distributed Unit as defined by O-RAN and having fronthaul split option 7-2x
O-DU$_x$	A specific O-RAN Distributed Unit having fronthaul split option x where x may be 6, 7-2x (as defined by WG4) or 8
O-RAN	Open-Radio Access Network
O-RU	O-RAN Radio Unit as defined by O-RAN and having fronthaul split option 7-2x
O-RU$_x$	A specific O-RAN Radio Unit having fronthaul split option x, where x is 6, 7-2x (as defined by WG4) or 8, and which is used in a configuration where the fronthaul interface is the same at the O-DU$_x$
O-RU$_y$	A specific O-RAN Radio Unit having fronthaul split option y, where y is 7-2x (as defined by WG4) or 8, and which is used in a configuration where the fronthaul interface at O-RU is different than that at O-DU$_x$
PDCP	Packet Data Conversion Protocol
POE	Power Over Ethernet
QoE	Quality of Experience
QoS	Quality of Service
RAN	Radio Access Network
RAT	Radio Access Technology
Rel-x	Release number: where x is the actual release number
RF	Radio Frequency
RFFE	RF Front End
RLC	Radio Link Control
RRC	Radio Resource Control
RRM	Radio Resource Management
RU	Radio Unit as defined by 3GPP
Rx	Receiver
SDAP	Service Data Adaptation Protocol
SDU	Service Data Unit
SON	Self Organized Network
SU-MIMO	Single User MIMO
TDD	Time division Duplex
TR	Technical Report
TRP	Transmission Reception Point
TS	Technical Specification
Tx	Transmitter
UE	User Equipment
UL	Uplink
UPF	User Plane Function
URLLC	Ultra-Reliable and Low Latency Communications
µsec	Microseconds
WG	Working Group

1.4 Objectives

Openness is one of the key principles for O-RAN Alliance. To take full advantage of the economies of scale offered by an open computing platform approach, O-RAN Alliance reference designs will specify high performance, spectral and energy efficient white-box base station hardware. Reference platforms support a decoupled approach and offer detailed schematics for hardware and software architecture to enable both the O-DU$_x$ and O-RU$_x$.

The promotion of white box hardware is a potential way to reduce the cost of 5G deployment, which will benefit both the operators and vendors. The objective of the white-box hardware workgroup is to specify and release a complete reference design, thereby fostering a decoupled software and hardware platform. Currently, there is no open base station reference design architecture, making it impossible for operators and vendors to develop software for optimizing their network operation in various application scenarios. Therefore, it is further envisioned that the group will do research on all related content to build valuable reference designs.

The main objective of this document is to aid in development of base station HW architecture and requirements specification, and consequently, the HW reference design specification for white boxes based on the identified deployment scenarios within this specification.

Chapter 2 Deployment Scenarios

2.1 General

In this chapter we present the deployment scenarios as identified by the operator group within O-RAN. The detailed reference base station specification for each identified scenario and their classifications case will be provided in the following chapters. The use case scenarios will include:

- eMBB (enhanced Mobile BroadBand)
- URLLC (Ultra-Reliable and Low Latency Communications)

2.1.1 Base Station Architecture: Background

For any deployment scenarios, the base station architecture falls into three categories: 1) Split Architecture, 2) Integrated architecture or 3) All-in-one architecture. The split architecture can be deployed in two ways: 1) O-CU and O-DU$_x$ are connected via an optional fronthaul multiplexer (FHM) or a switch to multiple O-RU$_x$ /optional cascaded O-RU$_x$ (See Figure 2.1.1-1 or Figure 2.1.1-2) O-CU and O-DU$_x$ are connected via a fronthaul gateway (FHGW$_{x\rightarrow y}$) to multiple O-RU$_y$s (See Figure 2.1.1-2). Note that the term optional is used to indicate there may be a deployment case where an FHM, or a switch, or a FHGW may not be used for a specific deployment scenario. The O-CU shall be located at the data center and the O-DU can be placed either at the data center or at the cell site. O-DUs located at the cell site can also be placed in a ruggedized cabinet for outdoor deployments.

Depending on how much intelligence one requires to place in O-RU$_x$ or O-DU$_x$, there will be a functional split that splits up functions of PHY layer between O-DU$_x$ and O-RU$_x$. O-CU and O-DU$_x$ may be designed to be integrated into one hardware box or as two separate boxes. For downlink, depending on the deployed architecture, FHM or FHGW broadcasts the data to all the O-RU$_x$s or O-RU$_y$s, respectively. For uplink, it shall combine all the uplink data from subtending O-RU$_x$s or O-RU$_y$s according to the cell sets. Within working group 7 (and therefore all specifications stemming from it), all fronthaul interfaces are based on the following three criteria. A fronthaul interface that meets one of the following criteria can be defined as an open fronthaul interface:

1. O-RAN WG4 (open fronthaul interface group) released interfaces; or
2. O-RAN approved publicly (e.g., small cell forum or etc.) available external interfaces; or
3. Fronthaul interfaces made available and published as part of an O-RAN approved WG7 reference design.

Within this specification, "dashed boxes" are representative of white boxes. In some cases, the figures use specific O-DU$_x$ and O-RU$_y$ or O-DU$_x$ and O-RU$_x$ terminologies to demonstrate an example split option configuration. The first case is where the FHGW$_{x\rightarrow y}$ translates split option x from an O-DU$_x$ to split option y for an O-RU$_y$. Currently, the following split options with a fronthaul translation and the associated terminologies will be used throughout all WG7 specifications:

1- O-DU$_{7\text{-}2x}$$\leftrightarrow$FHGW$_{7\text{-}2x\rightarrow 8}$$\leftrightarrow$ O-RU$_8$

In the case where the fronthaul gateway does not perform a protocol translation, the terminology FHM O-DU$_x$ and O-RU$_x$ are used, and the following split options and the associated terminologies will be used throughout all specifications:

1- O-DU$_6$ $\leftrightarrow$ FHM/Switch$\leftrightarrow$ O-RU$_6$
2- O-DU$_{7\text{-}2x}$ $\leftrightarrow$ FHM/Switch$\leftrightarrow$ O-RU$_{7\text{-}2x}$
3- O-DU$_8$ $\leftrightarrow$ FHM/Switch$\leftrightarrow$ O-RU$_8$

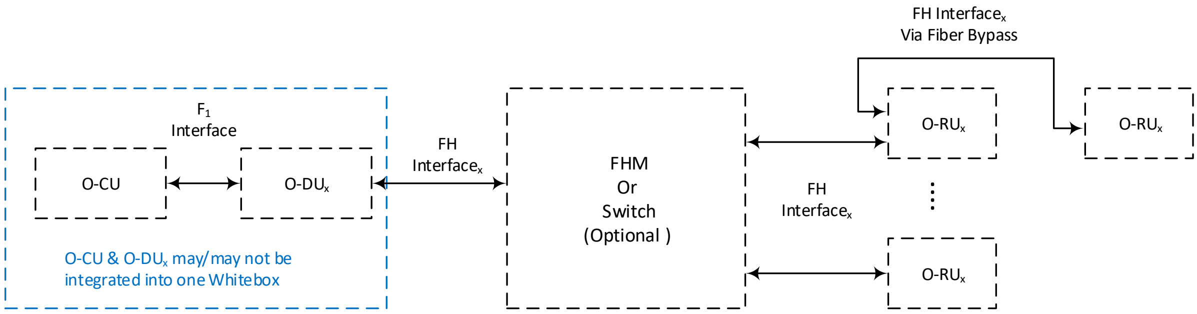

Figure 2.1.1-1: Base Station Architecture deploying lower level split without protocol translation

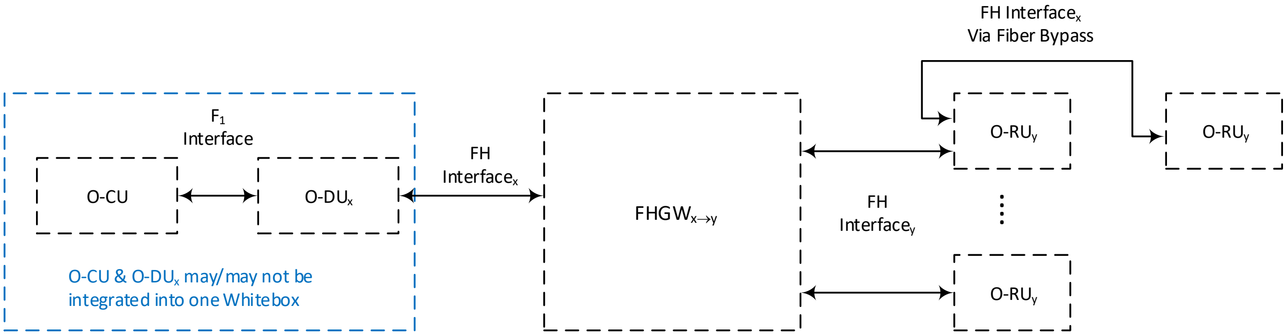

Figure 2.1.1-2: Base Station Architecture deploying lower level split with protocol translation

Note that within WG7 and therefore all specifications stemming from it, O-DU$_{7\text{-}2x}$ and O-RU$_{7\text{-}2x}$ refer to O-DU and O-RU as defined by WG4.

Since in split architecture O-DU$_x$ and O-RU$_x$ are physically separated from one another by the FH interface, therefore, depending on the split option, certain functions of base stations may either be performed in the O-DU$_x$ or O-RU$_x$. There are 8 different FH split options available (as defined by 3GPP); however, currently within WG7 the following split options are considered (as requested by operators for their deployment scenarios):

1. Split option 6: All PHY functions will reside in O-RU$_x$ while MAC functions will be performed in O-DU$_x$. In other words, only un-coded user data is on FH. In this case the terminology O-DU$_6$ and O-RU$_6$ is used.
2. Split option 7-2x, the PHY is split into High and Low PHY functions; where High PHY functions (coding, rate matching, scrambling, modulations and layer mapping) are performed in O-DU$_x$ while O-RU$_x$ performs the Low PHY functions (precoding, remapping, digital beamforming, IFFT and CP insertion). All I/Q samples are in frequency domain. In this case the terminology O-DU$_{7\text{-}2x}$ and O-RU$_{7\text{-}2x}$ is used.
3. Split Option 8: All PHY functions are performed in O-DU$_x$. This means that O-RU$_x$ function is limited to RF to baseband conversion and vice versa. The I/Q samples on FH interface are in time-domain. In this case the terminology O-DU$_8$ and O-RU$_8$ is used.

In some cases, when a FHM/Switch/FHGW is not used, a second O-RU may be cascaded to the first O-RU via the FH interface. In this way, the coverage area may be further expanded. The FH interface between the two cascaded O-RUs shall be the same as the FH interface between the O-DU and the first O-RU. For downlink, the first O-RU copies the data received from the O-DU, then sends it to the second O-RU. For uplink, the first O-RU combines the uplink data from the second O-RU, and then sends it to the O-DU.

Integrated Architecture is defined as where O-DU$_x$ and O-RU$_x$ are physically located in one hardware box and as such no FH interface is required to connect them. Figure 2.1.1-3 illustrates the base station having an integrated architecture. For the integrated architecture, open F1 interface is between O-CU and O-DU$_x$ (which is discussed in O-RAN WG5).

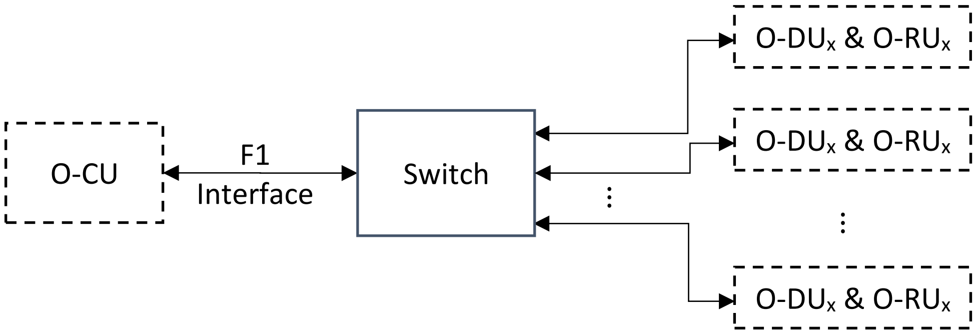

Figure 2.1.1-3: Base Station with Integrated Architecture

All-in-one Architecture is defined as where O-CU, O-DU$_x$ and O-RU$_x$ are physically located in one hardware box, as such no FH interface and F1 interface are required to connect them. Figure 2.1.1-4 illustrates the base station having an all-in-one architecture. The functions of O-CU, O-DU$_x$ and O-RU$_x$ are included in baseband unit followed by RF Processing unit.

Figure 2.1.1-4: Base Station with All-in-one Architecture

2.2 Deployment Scenarios

Deployment scenarios for use-cases described in section 2.1 will be provided in this section. These scenarios are based on reported scenarios as reported by service providers. Scenarios in this specification do not preclude any other scenario. Other scenarios may be added as a change request to this specification.

2.2.1 Indoor Picocell

The indoor deployment scenario is mainly for small coverage per site within buildings, open spaces, etc. A key characteristic of this deployment scenario is consistent indoor user experience, high user density within buildings an open office and high capacity.

Some of its attributes are listed in Table 2.2.1-1 below.

Table 2.2.1-1: Attributes for FR1 Indoor Picocell

Attributes	Values or Assumptions
Carrier Frequency [(a)]	n2, n4, n5, n13, n41, n48, n66, n77, n78, n79
System Bandwidth[(b)]	Up to 100MHz (DL+UL)
Cell Layout	Single layer
Mounting options	Above ceiling (Plenum rated) and/or on the wall; Open Office
ISD	20m
BS MIMO Configuration[(c)]	2TX/2RX or 4TX/4RX
Output power	2TX/2RX: ≤27dBm (per channel) Conducted Output Power; 4TX/4RX: ≤24dBm (per channel) Conducted Output Power
Coverage[(d)]	Omni
Key Architecture Features	Split Architecture (split options: 6, 7-2x and 8 are supported and each option is further described in Section 2.1.1) and Integrated Architecture.

Notes:

(a) The options noted here are for the first stage, and do not preclude the study of other spectrum options, e.g. mmWave. Certain technology bands have specific regulatory requirements that must be followed

(b) The system bandwidth is the total bandwidth typically assumed to derive the values for some KPIs such as area traffic capacity and user experienced data rate. It is also allowed to have an aggregated bandwidth totalling up to the system bandwidth. "DL + UL" refers to either of the following two cases:

- FDD with symmetric bandwidth allocations between DL and UL.
- TDD with the aggregated system bandwidth used for either DL or UL via switching in time-domain.

(c) The options noted here are typical configurations.

(d) Other options (e.g. Sectorized with directional antenna), are not precluded.

2.2.2 Outdoor Picocell

Two scenarios have been identified for picocell outdoor scenario: 1) FR1 and 2) FR2. While both deployment scenarios have integrated architecture (See Figure 2.1.1-3), they do have different attributes and as such they have been listed separately in Table 2.2.2-1 and Table 2.2.2-2.

Table 2.2.2-1: Attributes for FR1 Outdoor Picocell

Attributes	Values or Assumptions
Carrier Frequency[(a)]	n2, n4, n5, n13, n41, n48, n66, n77, n78, n79
System Bandwidth[(b)]	≥100 MHz (DL+UL)
Cell Layout	single layer Two layers: - Macro or Micro layer - Pico layer: random locations
Mounting Options	Rooftop, Side of Building (wall), Pole, under overhang
ISD	200 m
BS MIMO Configuration[(c)]	Up to 4TX/4RX
Output Power	≥35 dBm EIRP
Coverage[(d)]	Omni
Key Architecture Features	Integrated Architecture. See Figure 2.1.1-3.

Notes:

(a) The options noted here are for the first stage, and do not preclude the study of other spectrum options, e.g. mmWave. Certain technology bands have specific regulatory requirements that must be followed

(b) The system bandwidth is the total bandwidth typically assumed to derive the values for some KPIs such as area traffic capacity and user experienced data rate. It is also allowed to have an aggregated bandwidth totalling up to the system bandwidth. "DL + UL" refers to either of the following two cases:

- FDD with symmetric bandwidth allocations between DL and UL.
- TDD with the aggregated system bandwidth used for either DL or UL via switching in time-domain.

(c) The options noted here are typical configurations.

(d) Other options (e.g., Sectorized with directional antenna), are not precluded.

Table 2.2.2-2: Attributes for FR2 Outdoor Picocell

Attributes	Values or Assumptions
Carrier Frequency[(a)]	n257, n258, n260, n261
System Bandwidth[(b)]	up to 800 MHz (DL+ UL)
Layout	single layer Two layers: - Macro or Micro layer - Pico layer: random locations
Mounting Options	Rooftop, Side of Building (wall), Pole, under overhang
ISD	200 m
BS MIMO Configuration[(c)]	Up to 4TX/4RX
Output Power	50-55 dBm EIRP
Coverage[(d)]	Omni
Key Architecture Features	Integrated Architecture. See Figure 2.1.1-3 .

Notes:

(a) The options noted here are for the first stage, and do not preclude the study of other spectrum options, e.g. mmWave. Certain technology bands have specific regulatory requirements that must be followed

(b) The system bandwidth is the total bandwidth typically assumed to derive the values for some KPIs such as area traffic capacity and user experienced data rate. It is also allowed to have an aggregated bandwidth totalling up to the system bandwidth. "DL + UL" refers to either of the following two cases:

- FDD with symmetric bandwidth allocations between DL and UL.
- TDD with the aggregated system bandwidth used for either DL or UL via switching in time-domain.

(c) The options noted here are typical configurations.

(d) Other options (e.g. Sectorized with directional antenna) are not precluded.

2.2.3 Outdoor Microcell

There are two deployment scenario proposals for outdoor microcell by operators' group. The architectures for both microcell deployment scenarios are the same and it is illustrated in Figure 2.1.1-1 and Figure 2.1.1-2, except that outdoor microcell have all-in-one architecture (See Figure 2.1.1-4).

The base station attributes for FR1 are listed in Table 2.2.3-1. The attributes for FR2 microcell outdoor deployment scenario are listed in Table 2.2.3-2. It shall be noted that all frequencies are given as NR operating bands in either in FR1 or FR2.

Table 2.2.3-1: Attributes for FR1 Outdoor Microcell

Attributes	Values or Assumptions	
	Outdoor Microcell	Enterprise Microcell

Carrier Frequency[a]	n2, n4, n5, n13, n41, n48, n66, n77, n78, n79	n41, n78
System Bandwidth[b]	Up to 200MHz (DL+UL)	Up to 200MHz (DL+UL)
Cell Layout	single layer Two layers: - Macro or Micro layer - Pico layer: random locations	single layer Two layers: - Macro or Micro layer - Pico layer: random locations
Mounting Options	Rooftop, Side of Building (wall), Pole	Rooftop, Side of Building (wall), Pole
ISD	700 m	500m (distance between the two cascaded O-RUs: Up to 10km)
BS MIMO Layers[c]	Up to 16TX/16RX	Up to 8TX/8RX
Output Power	< 61 dBm EIRP	< 64 dBm EIRP
Coverage[d]	Omni/Sectorized	Omni/Sectorized
Key Architecture Features	Split architecture (O-RAN WG4 split option 7-2x) and all-in-one architecture.	Split architecture (O-RAN WG4 split option 7-2x, or split option 8).

Notes:

(a) The options noted here are for the first stage, and do not preclude the study of other spectrum options, e.g., mmWave. Certain technology bands have specific regulatory requirements that must be followed

(b) The system bandwidth is the total bandwidth typically assumed to derive the values for some KPIs such as area traffic capacity and user experienced data rate. It is also allowed to have an aggregated bandwidth totalling up to the system bandwidth. "DL + UL" refers to either of the following two cases:

- FDD with symmetric bandwidth allocations between DL and UL.
- TDD with the aggregated system bandwidth used for either DL or UL via switching in time-domain.

(c) The options noted here are typical configurations.

(d) Other options (e.g. Sectorized with directional antenna) are not precluded.

Table 2.2.3-2: Attributes for FR2 Outdoor Microcell

Attributes	Values or Assumptions
Carrier Frequency[a]	n257, n258, n260, n261
System Bandwidth[b]	Up to 1.2 GHz (DL+UL)
Cell Layout	single layer Two layers: - Macro or Micro layer - Pico layer: random locations
Mounting Options	Rooftop, Side of Building (wall), Pole
ISD	200 m
BS MIMO Layers[c]	Up to 4TX/4RX
Output Power	>60 dBm EIRP
Coverage[d]	Sectorized
Key Architecture Features	Split Architecture (O-RAN WG4 split option 7-2x).

Notes:

(a) The options noted here are for the first stage, and do not preclude the study of other spectrum options, e.g., mmWave. Certain technology bands have specific regulatory requirements that must be followed

(b) The system bandwidth is the total bandwidth typically assumed to derive the values for some KPIs such as area traffic capacity and user experienced data rate. It is also allowed to have an aggregated bandwidth totalling up to the system bandwidth. "DL + UL" refers to either of the following two cases:

- FDD with symmetric bandwidth allocations between DL and UL.
- TDD with the aggregated system bandwidth used for either DL or UL via switching in time-domain.

(c) The options noted here are typical configurations.

(d) Other options (e.g. Sectorized with directional antenna) are not precluded.

2.2.4 Integrated access and backhaul (IAB)

A key benefit of Integrated Access and Backhaul (IAB) is enabling flexible and very dense deployment of NR cells without densifying the transport network proportionately. Various deployment scenarios have been proposed including support for outdoor small cell deployments, indoors, or even mobile relays (e.g., on buses or trains). The IAB architecture is shown in Figure 2.2.4-1below. IAB specifications as well as F1* interface can be found in [3].

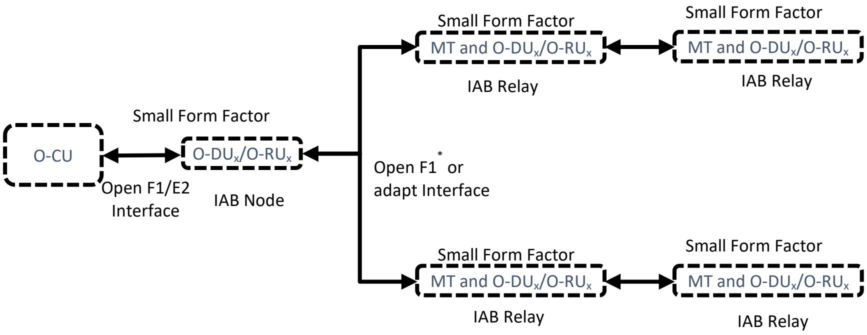

Figure 2.2.4-1: IAB integrated architecture.

Some of the attributes of integrated access and backhaul are listed in Table 2.2.4-1.

Table 2.2.4-1: Attributes for Integrated Access and Backhaul

Attributes	Values or assumptions
Carrier Frequency[a]	n257, n258, n260, n261
System Bandwidth[b]	n257, n258, n261: 100 – 400 MHz (DL+ UL), n260: 400 – 800 MHz (DL+ UL)
Cell Layout	Single or HetNet, single hop to multi-hop
Mounting Options	Rooftop, Side of Building (wall), Pole, under overhang, vehicle (bus)
ISD	200 m
BS MIMO Layers[c]	Up to 4TX/4RX
Output Power	50-55 dBm EIRP, to be confirmed after 3GPP RAN4 confirmation
Coverage [d]	Omni
Key Architecture Features[e]	Integrated Architecture (O-DU + MT functionality)

Notes:

(a) The options noted here is for the first stage, and do not preclude the study of other spectrum options.

(b) "DL + UL" refers to either of the following two cases:

- FDD with symmetric bandwidth allocations between DL and UL.
- TDD with the aggregated system bandwidth used for either DL or UL via switching in time-domain.

(c) The options noted here are typical configurations.

(d) Other options (e.g., Sectorized with directional antenna) are not precluded.

(e) MT is the UE function in each IAB node that allows it to communicate with parent node.

2.2.5 Outdoor Macrocell (OMAC)

The outdoor deployment scenario is mainly for macro coverage for roads, buildings, open spaces, etc. in rural, semi-urban and urban areas. A key characteristic of this deployment scenario is consistent 5G experience and seamless mobility.

The Macrocell configuration is defined below for split option 7-2x with the O-DU at the cell site or at the data center and can be an integrated architecture (Split 2 architecture – Integrated O-DU and O-RU).
Some of its attributes are listed in Table 2.2.5-1 below.

The outdoor massive MIMO radio unit deployment scenario is mainly used for improved coverage and capacity solution utilizing both spatial multiplexing and beamforming. It offers higher capacity and better user experience when compared to traditional 8T8R radio. Massive MIMO O-RU includes active antenna array system with higher number of radio RF chains. This can be deployed for high traffic roads, high rise buildings, open spaces, etc. and in rural, urban and dense urban areas.

The massive MIMO radio configuration is defined below for split option 7-2x with the O-DU at the cell site or at the data center. The massive MIMO specific attributes are listed in Table 2.2.5-1 below. Massive MIMO will have separate chapters in OMAC HAR and HRD documents to describe the high- and low-level architectures in detail, respectively

Table 2.2.5-1: Attributes for FR1 Outdoor Macrocell

Attributes	Values or Assumptions
Carrier Frequency[(a)]	n26, n29, n41, n66, n70, n71, n77, n78, B1/n1, B3/n3, B7/n7, n78 CEPT(e)).
System Bandwidth[(b)]	up to 400 MHz TDD; up to 205 MHz FDD
Cell Layout	up to 8T8R O-RU: up to 4-layer MIMO 16T16R to 64T64R Massive MIMO O-RU: up to 16-layer MIMO (DL) and; up to 12-layer MIMO (UL)
Mounting Options	Rooftop, Towers, Side of Buildings
ISD	Variable
BS MIMO Layers[(c)]	Up to 8T8R For Massive MIMO Radio Unit:16T16R to 64T64R
Output Power	Total Radiated Power: Variable (Default 40W/10MHz)
Coverage[(d)]	Sectorized, Directional, Beamforming (Fixed/Dynamic)
Key Architecture Features	Split Architecture (O-RAN WG4 split option 7-2x), see Figure 2.1.1-1: Base Station Architecture deploying lower level split without protocol translation Option for O-DU at Data Center or Cell Site; Integrated Architecture. See Figure 2.1.1-3: Base Station with Integrated Architecture

Notes:

(a) The options noted here are for the first stage, and do not preclude the study of other spectrum options, e.g., CBRS. Certain technology bands have specific regulatory requirements that must be followed

(b) The system bandwidth is the total bandwidth typically assumed to derive the values for some KPIs such as area traffic capacity and user experienced data rate.

(c) The options noted here are typical configurations.

(d) Other options (e.g. Omni antennas) are not precluded. .

(e) n78 CEPT makes reference to a European n78 sub-band (according with ECC Decision (11) 06 and CEPT

2.2.6 Fronthaul Gateway

A Fronthaul Gateway (FHGW) is deployed between an $O\text{-}DU_x$ and an $O\text{-}RU_y$ and provides the necessary protocol translation between splits x and y. Such a Fronthaul Gateway is being referred to as $FHGW_{x\rightarrow y}$. In Addition to the protocol translation capabilities, a Fronthaul Gateway can provide packet transport and aggregation capabilities.

A Fronthaul Gateway can be deployed in any of the scenarios mentioned in the sections above. Generally, the attributes of a Fronthaul Gateway are given in Table 2.2.6-1.

Table 2.2.6-1: Attributes for Fronthaul Gateway

Attributes	Values or Assumptions
Functionalities	Packet Switching, Radio Interface Translation
Protocol Translation Support	Translation between Split Option 8 and Split Option 7-2x (Refer to [6])
Timing Capabilities	Compliant to S-Plane Protocol Specification in [6]
Deployment	Indoor and Outdoor
Interfaces	Interfaces to connect to O-DU_x Interfaces to connect to O-RU_y

Chapter 3 Base Station Type Classification

3.1 General

The base stations for each usage scenario in chapter 2 apply to Wide Area Base Stations, Medium Range Base Stations and Local Area Base Stations. The definition for the three BS classes is given in subsection 4.4 of [2]. The BS classes for each deployment scenario described in this document are defined in Table 2.2.6-1 below.

Table 2.2.6-1: Base Station Classes for Deployment Scenarios

Deployment Scenario	FR1/FR2	Base Station Class
Indoor Picocell Split Architecture	FR1	Local Area/Medium Area
Indoor Picocell Integrated Architecture	FR1	Local Area/Medium Area
Outdoor Picocell Integrated Architecture	FR1	Medium Area
	FR2	Medium Area
Outdoor Microcell Split Architecture	FR1	Medium/Wide Area
	FR2	Medium/Wide Area
Outdoor Microcell All-in-one Architecture	FR1	Medium Area
Integrated Access and Backhaul	FR2	Medium Area
Outdoor Macro	FR1	Wide Area

3.2 Indoor

For indoor coverage, the base station can be identified as Local Area Base Station, and there are two key architecture types, namely, the split architecture and the integrated architecture.

NOTE: The hardware design may consider the co-location of MEC and UPF part in O-CU/O-DU functionality if needed.

3.2.1 The split architecture

Descriptions of the indoor split architecture are listed in Table 3.2.1-1.

Table 3.2.1-1: Descriptions of Attributes for the Indoor Picocell Split Architecture

Attributes	Description
Reference Figure	Figure 2.1.1-1: Base Station Architecture deploying lower level split without protocol translation
Interface[a]	Open fronthaul interface
Functionality [b]	Split Option 6: O-RU$_6$: RF, IF, PHY (Downlink and uplink digital processing, baseband signal conversion to/from RF signal) O-CU & O-DU$_6$: MAC, RLC, PDCP (Downlink and uplink baseband processing, supply system synchronization clock, signalling processing, OM function, interface with core network) Split Option 7-2x: O-RU: RF, IF, PHY-LOW (Downlink and uplink digital processing, baseband signal conversion to/from RF signal) O-CU & O-DU: PHY-HIGH, MAC, RLC, PDCP (Downlink and uplink baseband processing, supply system synchronization clock, signalling processing, OM function, interface with core network) Split Option 8: O-RU$_8$: Downlink baseband signal convert to RF signal, uplink RF signal convert to baseband signal, interface with the fronthaul gateway O-CU &O-DU$_8$: Downlink and uplink baseband processing, supply system synchronization clock, signalling processing, OM function, interface with core network and the fronthaul gateway Fronthaul gateway: Downlink broadcasting, uplink combining, power supply for O-RU$_8$, cascade with other fronthaul gateway, synchronization clock
Transmission media[b]	From O-DU to Fronthaul gateway: Fiber; From Fronthaul gateway to O-RU: Fiber or Optical Electric Composite Cable From (O-DU&O-CU) to O-RU: Fiber or Optical Electric Composite Cable From (O-DU&O-CU) and core network: Fiber or Optical Electric Composite Cable; Trusted or Untrusted transport
Duplex[b]	TDD, FDD
Deployment location	Indoor
Cell Coverage	Omni
MIMO Configuration[b]	SU-MIMO/MU-MIMO; up to 4TX/4RX
User distribution[c]	20-200
Fronthaul Latency	Split Option 6: TBD Split Option 7-2x: fronthaul dependent (refer to [6]) Split Option 8: 150us
Synchronization[b]	GNSS, IEEE-1588v2, BDS
Power Supply Mode[b]	POE or by optical electric composite cables

Notes:

(a) The option noted here is for the first stage, and do not preclude other interface options.
(b) The options noted here are for reference, and do not preclude other options.
(c) The number of users distribution refers to the active users in one base station, and the options are for reference.

3.2.2 The integrated architecture

Descriptions of the integrated architecture are listed in Table 3.2.2-1.

Table 3.2.2-1: Descriptions of Attributes for the Indoor Picocell Integrated Architecture

Attributes	Description
Reference Figure	Figure 2.1.1-3
Interface[a]	F1
Functionality[b]	O-DU+O-RU: Downlink and uplink baseband processing, supply system synchronization clock, signalling processing, OM function, interface with core network and, downlink baseband signal convert to RF signal, uplink RF signal convert to baseband signal
Transmission media[b]	Fiber or Optical Electric Composite Cable
Duplex[b]	TDD, FDD
Deployment location	Indoor
Cell Coverage	Omni
MIMO Configuration[b]	SU-MIMO/MU-MIMO; up to 4TX/4RX
User distribution[c]	100
Fronthaul Latency	Not Applicable
Synchronization[b]	GNSS, IEEE-1588v2, Sniffer, BDS
Power Supply Mode[b]	POE or by optical electric composite cables

Notes:

(a) The option noted here is for the first stage, and do not preclude other interface options.
(b) The options noted here are for reference, and do not preclude other options.
(c) The number of users distribution refers to the active users in one base station, and the options are for reference

3.3 Outdoor

For the outdoor scenario, the base station can be identified as a Medium Range Base Station, and there are three potential key architecture types, which are the split architecture, the integrated architecture and the all-in-one architecture.

Note: The hardware design may consider the co-location of MEC and UPF part in O-CU/O-DU functionality if needed.

3.3.1 Picocell

The base station architecture for Picocell incorporates an integrated O-DU/O-RU architecture, and its description is listed in Table 3.3.1-1.

Table 3.3.1-1: Descriptions of Attributes for the Outdoor Picocell Architecture

Attributes	Description
Reference Figure	Figure 2.1.1-3
Interface[a]	F1
Functionality[b]	O-DU+O-RU: Downlink and uplink baseband processing, supply system synchronization clock, signalling processing, OM function, interface with core network and, downlink baseband signal convert to RF signal, uplink RF signal convert to baseband signal
Transmission media[b]	Fiber or Optical Electric Composite Cable
Duplex[b]	TDD
Deployment location	Outdoor
Cell Coverage	Omni
MIMO Configuration[b]	SU-MIMO/MU-MIMO; up to 4TX/4RX
User distribution[c]	>200
Fronthaul Latency	Not Applicable
Synchronization[b]	GNSS or IEEE-1588v2
Power Supply Mode[b]	POE or by optical electric composite cables

Notes:

(a) The option noted here is for the first stage, and do not preclude other interface options.
(b) The options noted here are for reference, and do not preclude other options.
(c) The number of users distribution refers to the active users in one base station, and the options are for reference

3.3.2 Microcell

The base stations for FR1 microcell scenarios incorporate potential key architecture types, which are the split architecture and the all-in-one architecture. The base stations for FR2 Outdoor microcell scenarios incorporate a split RAN architecture. Their descriptions are listed in Table 3.3.2-1, Table 3.3.2-2, Table 3.3.2-3 and Table 3.3.2-4

Table 3.3.2-1: Descriptions of Attributes for the FR1 Outdoor Microcell Architecture

Attributes	Description
Reference Figure	Figure 2.1.1-1: Base Station Architecture deploying lower level split without protocol translation Figure 2.1.1-2
Interface[a]	Open fronthaul interface (WG4) (Option 7-2x)
Functionality[b]	Provides lower layer split in RAN so that PHY-Hi, MAC/RLC/PDCP/SDAP/RRC could be centralized or virtualized and support CoMP features etc.
Transmission media[b]	Fiber, wireless to fiber
Duplex[b]	TDD
Deployment location	Outdoor
Cell Coverage	Omni; Sectorized
MIMO[b]	SU-MIMO/MU-MIMO; up to 16TX/16RX
User distribution[c]	>200
Fronthaul Latency	Split Option 7-2x: fronthaul dependent (refer to [6])
Synchronization[b]	GNSS or IEEE-1588v2
Power Supply Mode[b]	DC/AC

Notes:

(a) The option noted here is for the first stage, and do not preclude other interface options.
(b) The options noted here are for reference, and do not preclude other options.
(c) The number of users distribution refers to the active users in one base station, and the options are for reference

Table 3.3.2-2: Descriptions of Attributes for the FR1 Enterprise Microcell Architecture

Attributes	Description
Reference Figure	Figure 2.1.1-1: Base Station Architecture deploying lower level split without protocol translation
Interface[(a)]	Open fronthaul interface
Functionality[(b)]	Split Option 7-2x: O-RU: RF, IF, PHY-LOW (Downlink and uplink digital processing, baseband signal conversion to/from RF signal) O-CU & O-DU: PHY-HIGH, MAC, RLC, PDCP (Downlink and uplink baseband processing, supply system synchronization clock, signalling processing, OM function, interface with core network) Split Option 8: O-RU: Downlink baseband signal convert to RF signal, uplink RF signal convert to baseband signal, interface with O-DU O-CU &O-DU: Downlink and uplink baseband processing, supply system synchronization clock, signalling processing, OM function, interface with core network and O-RU.
Transmission media[(b)]	Fiber
Duplex[(b)]	TDD
Deployment location	Outdoor
Cell Coverage	Omni; Sectorized
MIMO[(b)]	SU-MIMO/MU-MIMO; up to 8TX/8RX
User distribution[(c)]	>300
Fronthaul Latency	Split Option 7-2x: fronthaul dependent (refer to [6]) Split Option 8: 150us
Synchronization[(b)]	GNSS or IEEE-1588v2
Power Supply Mode[(b)]	DC/AC

Notes:

(a) The option noted here is for the first stage, and do not preclude other interface options.
(b) The options noted here are for reference, and do not preclude other options.
(c) The number of users distribution refers to the active users in one base station, and the options are for reference

Table 3.3.2-3: Descriptions of Attributes for the FR1 Outdoor Microcell with All-In-One Architecture

Attributes	Description
Reference Figure	Figure 2.1.1-1: Base Station Architecture deploying lower level split without protocol translation
Interface	Not Applicable
Functionality(a)	O-CU+O-DU+O-RU: RF, IF, PHY-LOW (Downlink and uplink digital processing, baseband signal conversion to/from RF signal), PHY-HIGH, MAC, RLC, PDCP (Downlink and uplink baseband processing, supply system synchronization clock, signalling processing, OM function, interface with core network)
Transmission media(a)	Fiber, wireless to fiber
Duplex(a)	TDD
Deployment location	Outdoor
Cell Coverage	Omni; Sectorized
MIMO(a)	SU-MIMO/MU-MIMO; up to 2TX/2RX
User distribution(b)	>300
Fronthaul Latency	Not Applicable
Synchronization(a)	GNSS or IEEE-1588v2
Power Supply Mode(a)	DC/AC

Notes:

(a) The option noted here is for reference, and do not preclude other options.
(b) The number of users distribution refers to the active users in one base station, and the options are for reference.

Table 3.3.2-4: Descriptions of Attributes for the FR2 Outdoor Microcell Architecture

Attributes	Description
Reference Figure	Figure 2.1.1-1: Base Station Architecture deploying lower level split without protocol translation
Interface(a)	F1 interface between O-CU/O-DU (Split Option 2) Open fronthaul interface (WG4) between O-DU/O-RU (Option 7-2x)
Functionality(b)	O-DU/O-RU: Downlink and uplink baseband processing, supply system synchronization clock, scheduling, signalling processing, OM function, interface with core network and, Downlink baseband signal convert to RF signal, uplink RF signal convert to baseband signal
Transmission media(b)	Fiber, wireless to fiber
Duplex(b)	TDD
Deployment location	Small cell, Outdoor, Urban/Suburban
Cell Coverage	Sectorized
MIMO Configuration(b)	SU-MIMO/MU-MIMO; up to 4TX/4RX
User distribution(c)	>100
Fronthaul Latency	Split Option 7-2x: fronthaul dependent (refer to [6])
Synchronization(b)	GNSS or IEEE-1588v2
Power Supply Mode(b)	AC or POE or by optical electric composite cables

Notes:

(a) The option noted here is for the first stage, and do not preclude other interface options.
(b) The options noted here are for reference, and do not preclude other options.
(c) The number of users distribution refers to the active users in one base station, and the options are for reference

3.3.3 Integrated access and backhaul (IAB)

The IAB base station uses an integrated architecture, and its attributes are listed in Table 3.3.3-1.

Table 3.3.3-1: Descriptions of Attributes for the IAB Architecture

Attributes	Description
Reference Figure	Figure 2.2.4-1
Interface[a]	F1 Interface (WG5) (split option 2) Open fronthaul interface (WG4) (split option 7-2x)
Functionality[b]	Provides support for multi-hop relay per Rel. 16.
Transmission media[b]	Fiber, wireless to fiber
Duplex[b]	TDD
Deployment location	Outdoor mobile/fixed relays, Urban/Suburban
Cell Coverage	Omni or Sectorized
MIMO Configuration[b]	SU/MIMO and MU/MIMO; up to 4TX/4RX
User distribution[c]	30 - 40
Latency	4ms/hop for eMBB
Synchronization[b]	GNSS, or OTA per Rel-16
Power Supply Mode[b]	AC, POE, or by optical electric composite cables

Notes:

(a) The option noted here is for the first stage, and do not preclude other interface options.
(b) The options noted here are for reference, and do not preclude other options.
(c) The number of users distribution refers to the active users in one base station, and the options are for reference

3.3.4 Macrocell

The base stations for macrocell scenarios incorporate a split RAN architecture. Their descriptions are listed in Table 3.3.4-1.

Table 3.3.4-1: Descriptions of Attributes for the FR1 Outdoor Macrocell Architecture

Attributes	Description
Reference Figure	Figure 2.1.1-1
Interface[(a)]	Split Architecture (O-RAN WG4 split option 7-2x).
Functionality[(b)]	Split Option 7-2x: O-RU: RF, IF, PHY-LOW (Downlink and uplink digital processing, baseband signal conversion to/from RF signal) O-DU: PHY-HIGH, MAC, RLC, PDCP (Downlink and uplink baseband processing, supply system synchronization clock, signalling processing) O-CU: PDCP, SDAP, RRC (Control and User Plane separation, RRM function, interface with core network)
Transmission media[(b)]	Fiber, wireless to fiber
Duplex[(b)]	FDD, TDD
Deployment location	Outdoor
Cell Coverage	Sectorized, directional
MIMO[(b)]	Up to 16 layer MIMO; up to 64TX/64RX
User distribution[(c)]	>200
Fronthaul Latency	160-200 µsec
Synchronization[(b)]	GNSS, IEEE-1588v2, SyncE
Power Supply Mode[(b)]	DC/AC

Notes:

(a) The option noted here is for the first stage, and do not preclude other interface options.
(b) The options noted here are for reference, and do not preclude other options.
(c) The number of Users distribution refers to the active users in one base station, and the options are for reference

Chapter 4 Key performance indicators

4.1 Peak data rate

Peak data rate is the highest theoretical data rate which is the received data bits assuming error-free conditions assignable to a single mobile station, when all assignable radio resources for the corresponding link direction are utilized (i.e., excluding radio resources that are used for physical layer synchronization, reference signals or pilots, guard bands and guard times).

Indoor	Outdoor			
Pico	Pico	Micro	IAB	Macro
0.75-1.33Gbps (DL); 0.25-0.67Gbps (UL); NOTE1	FR1: up to 6/12 Gbps (UL/DL) FR2: up to 15/30 Gbps (UL/DL)	FR1: up to 6/12 Gbps (UL/DL) FR2: up to 15/30 Gbps (UL/DL)	FR2: 3-12 Gbps (DL); 1.5-6 Gbps (UL) For n260: 12-24 Gbps (DL); 6-12 Gbps (UL)	FR1: up to 6/12 Gbps (UL/DL)

NOTE1: This requirement is calculated based on the frame structure UL:DL=1:1 (TDD).

4.2 Peak spectral efficiency

Peak spectral efficiency is the highest theoretical data rate (normalized by bandwidth), which is the received data bits assuming error-free conditions assignable to a single mobile station, when all assignable radio resources for the corresponding link direction are utilized (i.e., excluding radio resources that are used for physical layer synchronization, reference signals or pilots, guard bands and guard times).

Higher frequency bands could have higher bandwidth, but lower spectral efficiency and lower frequency bands could have lower bandwidth but higher spectral efficiency. Thus, peak data rate cannot be directly derived from peak spectral efficiency and bandwidth multiplication.

Peak spectral efficiency (bps/Hz) = Peak data rate (bps)/ bandwidth (Hz).

Indoor	Outdoor			
Pico	Pico	Micro	IAB	Macro
7.5-13.3 bps/Hz (DL); 2.5-6.7 bps/Hz (UP); NOTE1	30 bps/Hz (DL) 15 bps/Hz (UL) NOTE1	30 bps/Hz (DL) 15 bps/Hz (UL) NOTE1	30 bps/Hz (DL) 15 bps/Hz (UL) NOTE1	30 bps/Hz (DL) 15 bps/Hz (UL)

NOTE1: This requirement is calculated based on the frame structure UL:DL=1:1 (TDD).

4.3 Bandwidth

Bandwidth means the maximal aggregated total system bandwidth. It may be supported by single or multiple RF carriers. It is a quantitative KPI.

Indoor	Outdoor			
Pico	Pico	Micro	IAB	Macro
100 MHz	FR1: up to 200 MHz FR2: up to 800 MHz	FR1: up to 200 MHz FR2: up to 1.2 GHz	FR2: 100 – 400 MHz For n260: 400 – 800 MHz	FR1: up to 200 MHz

4.4 Control plane latency

Control plane latency refers to the time for UE to move from a battery efficient state (e.g., IDLE) to start of continuous data transfer (e.g., ACTIVE).

Indoor	Outdoor			
Pico	Pico	Micro	IAB	Macro
10ms	10ms	10ms	10ms/hop	10ms

4.5 User plane latency

The time it takes to successfully deliver an application layer packet/message from the radio protocol layer 2/3 SDU ingress point to the radio protocol layer 2/3 SDU egress point via the radio interface in both uplink and downlink directions, where neither device nor base station reception is restricted by DRX.

Indoor	Outdoor			
Pico	Pico	Micro	IAB	Macro
URLLC UL/DL(ms): 0.5ms eMBB: UL/DL(ms): 4ms	URLLC UL/DL(ms): 0.5ms eMBB: UL/DL(ms): 4ms	URLLC UL/DL(ms): 0.5ms eMBB: UL/DL(ms): 4ms	URLLC UL/DL(ms): 0.5ms/hop eMBB: UL/DL(ms): 4ms/hop	URLLC UL/DL (ms): 1ms eMBB: UL/DL (ms): 4ms

4.6 Mobility

Mobility means the maximum user speed at which a defined QoS can be achieved (in km/h).

Indoor	Outdoor			
Pico	Pico	Micro	IAB	Macro
3km/h	<60km/h	<500km/h (high speed train)	120 km/h	120 km/h (excluding high speed train)

